Linear Regression Models

Multi-variable models

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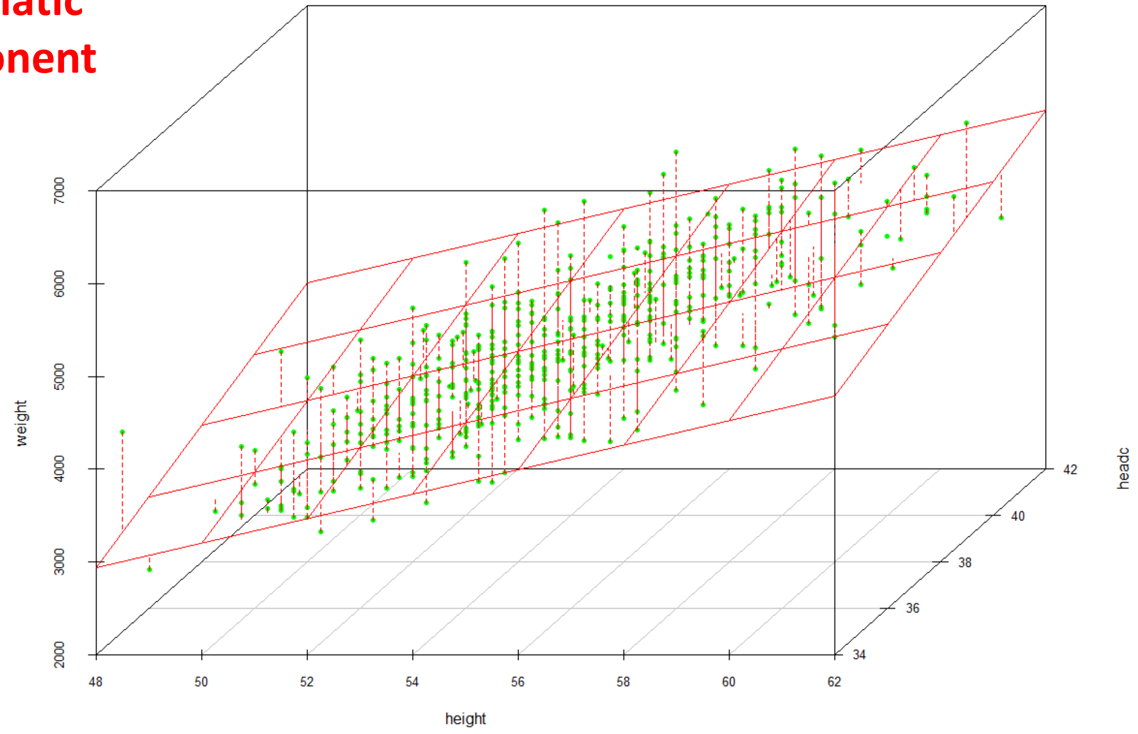
Multiple Linear regression

$$y_i = \boxed{\beta_o + \beta_1 * x_{1i} + \beta_2 * x_{2i} + \beta_3 * x_{3i} + \ldots + \beta_p * x_{pi}} + \varepsilon_i$$

Systematic component

Random error



Two explanatory variables: a *plane* in three-dimensional space

Model building: Possible strategies

– **sample size calculation** (at least 10-15 participants per variable; 50 + 8k, where k is the number of variables; 104 + k)

– **existing knowledge** (e.g., confounders, moderation effects) should be used

– models should be **interpretable**

Possible strategies

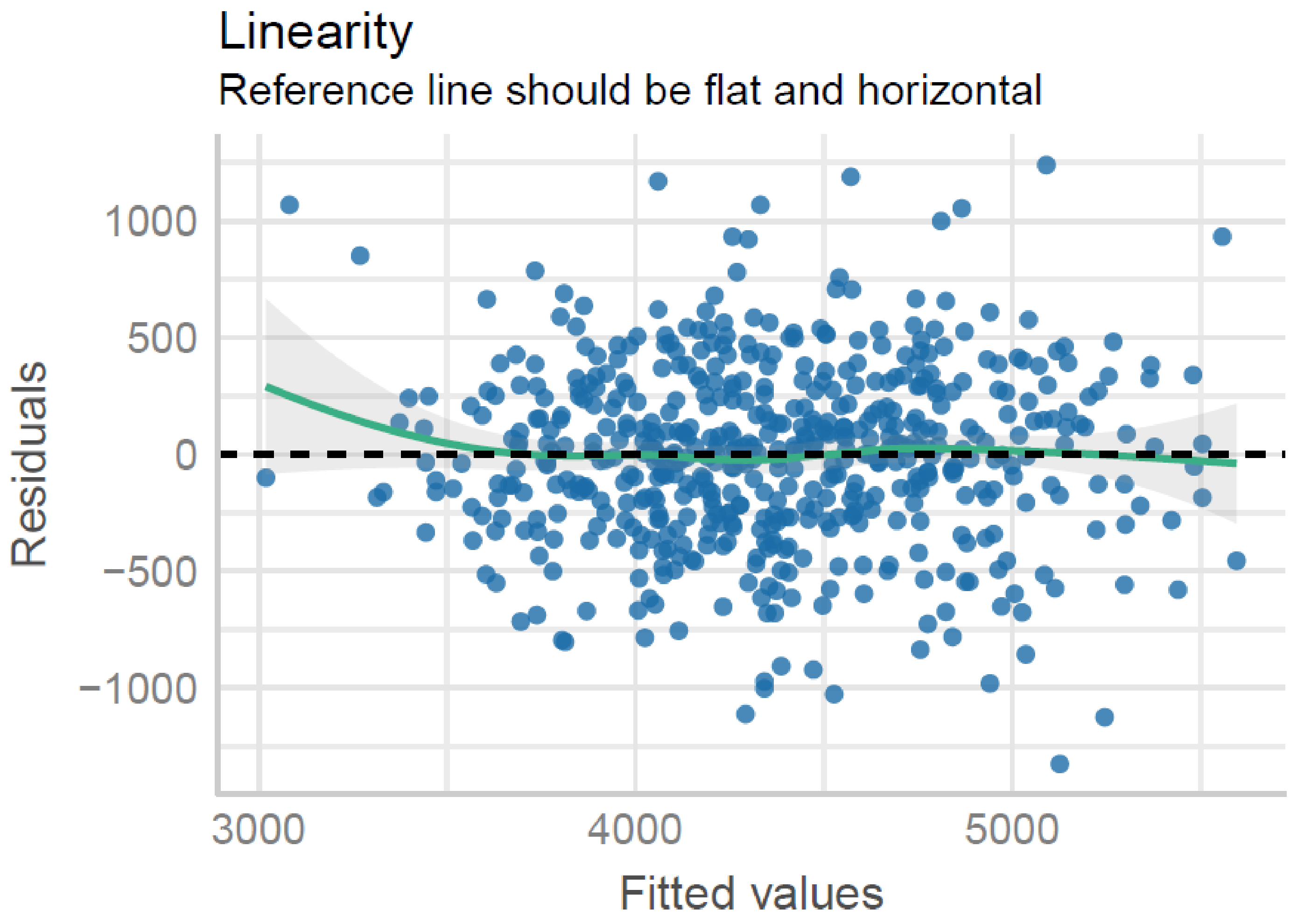
- Simultaneous Regression
- Hierarchical Regression (blocks)
- Purposeful selection process (e.g., the variables that have a p-value <0.2 in the univariable analysis are candidate variables for the model)
- Automated regression methods (using backward elimination, forward selection, or both[stepwise selection])

Simultaneous Regression

First, we will include all available explanatory variables and assess model fit based on this complete model.

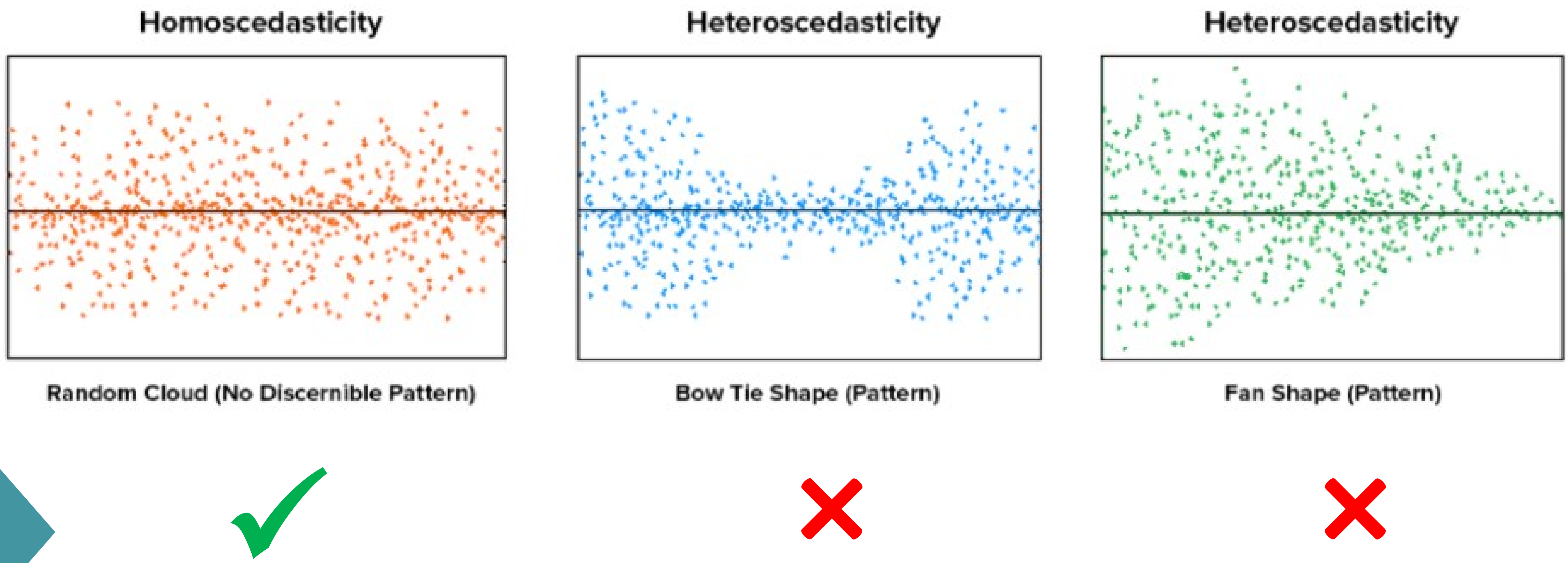
```
model_full <- lm(weight ~ ., data = dat)
```

Checking assumptions: Linearity



If you find equally spread residuals around a **horizontal line** without distinct **patterns**, that is a good indication for linear association.

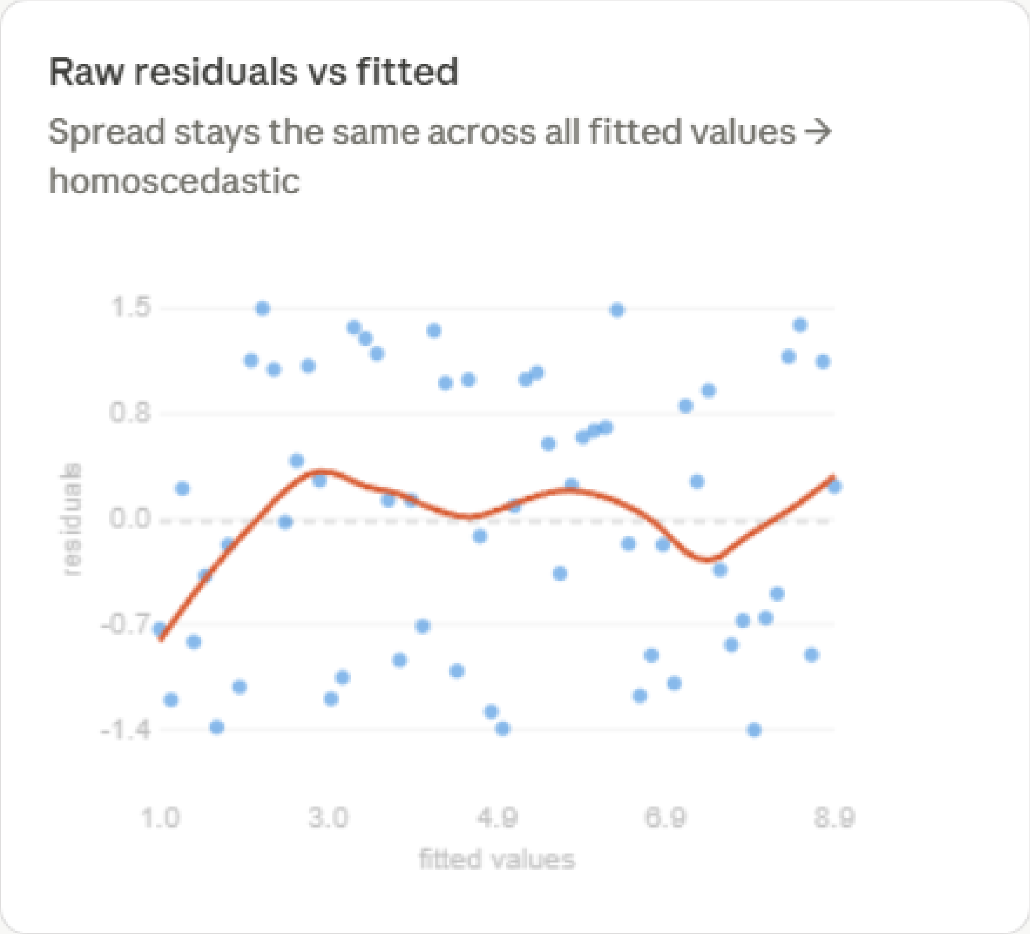
Checking assumptions: Homoscedasticity

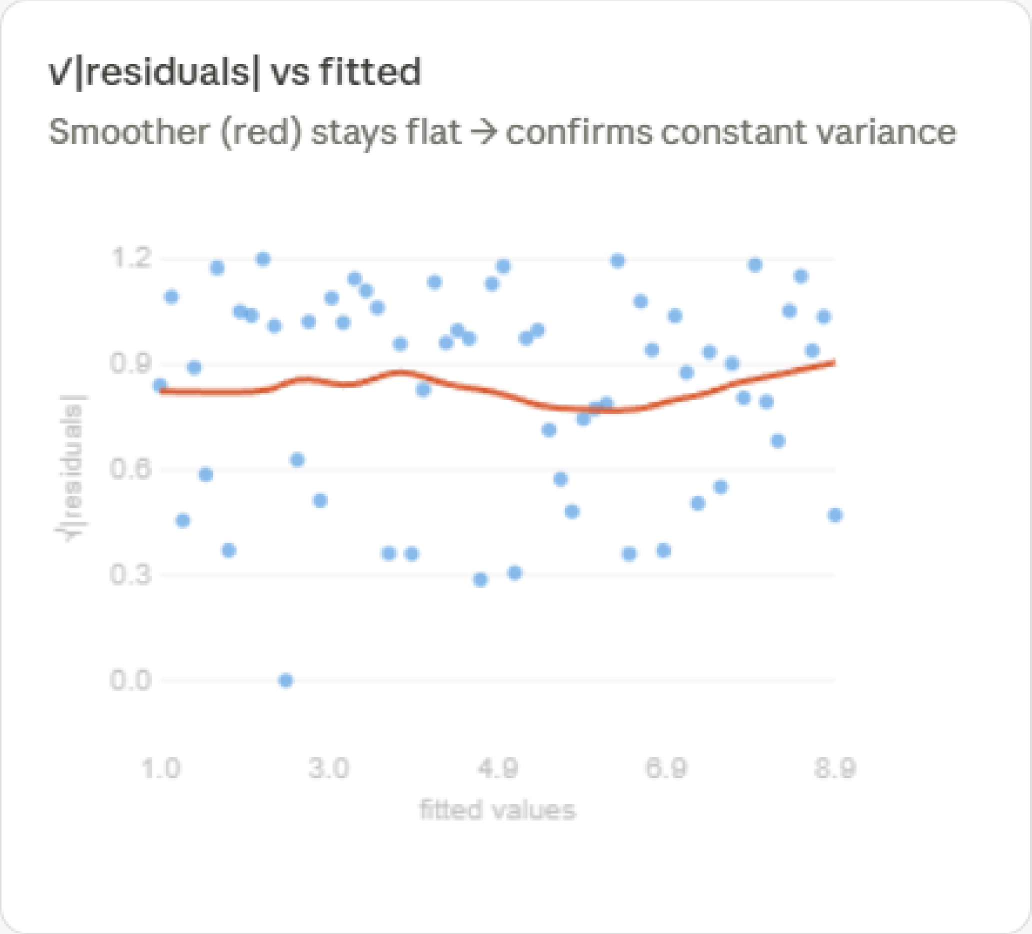


Example with Homoscedasticity

Y axis: values of the standardized residuals

Y axis: square root of the absolute values of the standardized residuals





residual point LOESS smoother

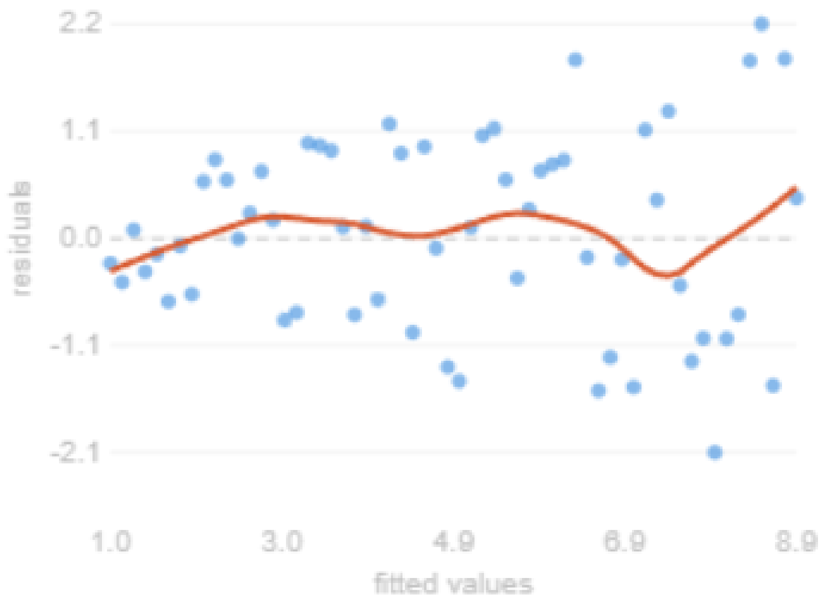
Example with Heteroscedasticity

Y axis: values of the standardized residuals

Y axis: square root of the absolute values of the standardized residuals

Raw residuals vs fitted

Fan shape: spread grows with fitted values → heteroscedastic

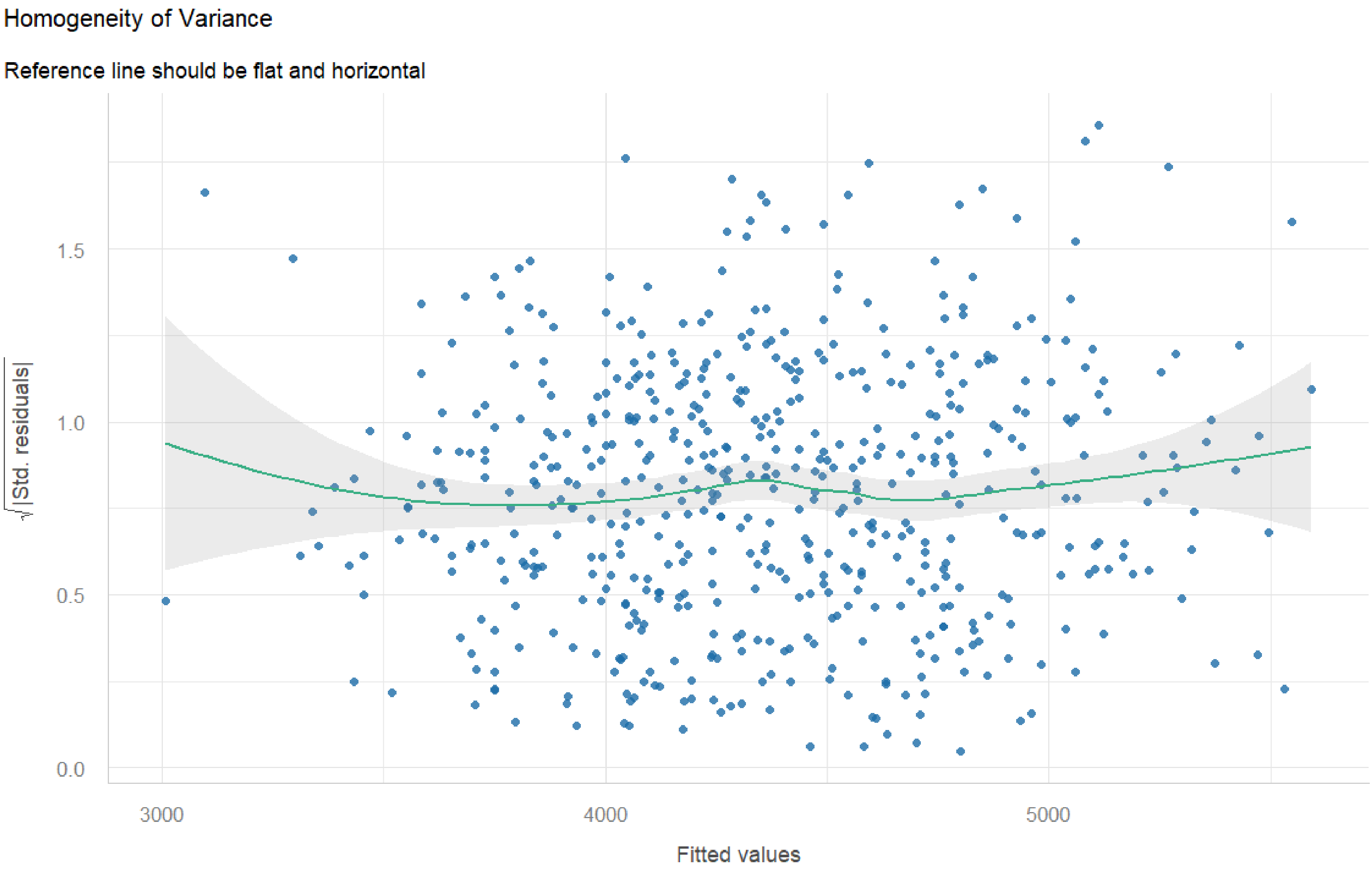


√|residuals| vs fitted

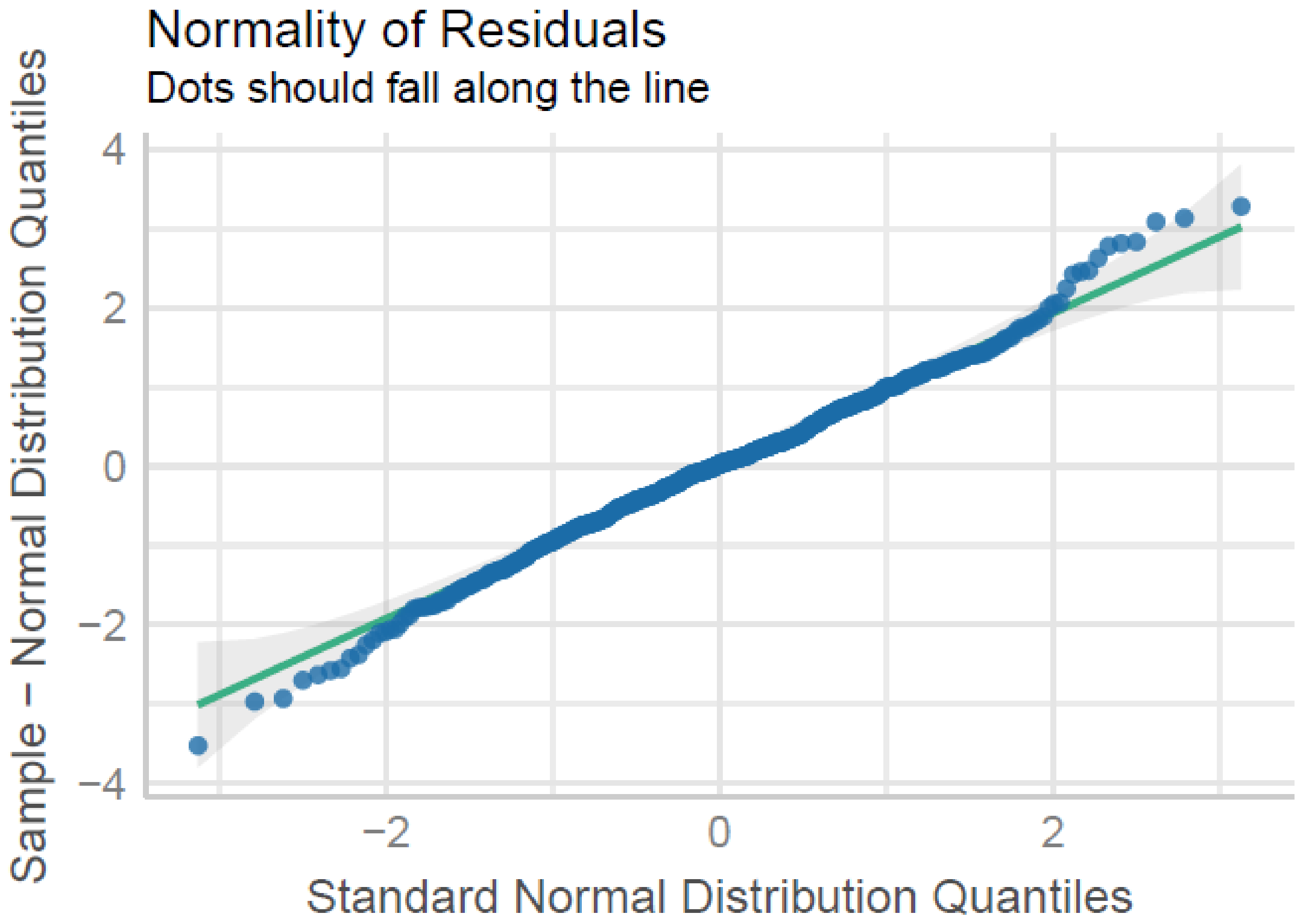
Smoother rises clearly → variance trend is now unmistakable

residual point LOESS smoother

Checking assumptions: Homoscedasticity



Normality of the residuals



Multicollinearity (between explanatory variables)

- Two or more **explanatory** variables are significantly related to one other, conditional on the other explanatory variables

- The amount of multicollinearity in a model can be estimated by the variance inflation factor (VIF).

VIF< 5 ➲ no-multicollinearity

Term	VIF	VIF 95% CI	adj. VIF	Tolerance	Tolerance 95% CI
height	1.60	[1.44, 1.81]	1.26	0.63	[0.55, 0.69]
headc	1.68	[1.51, 1.90]	1.29	0.60	[0.53, 0.66]
gender	1.14	[1.06, 1.29]	1.07	0.88	[0.77, 0.94]
education	1.07	[1.02, 1.27]	1.02	0.93	[0.79, 0.98]
parity	1.08	[1.02, 1.26]	1.02	0.93	[0.79, 0.98]

The smallest possible value for VIF is 1, which indicates the complete absence of collinearity. A VIF less than 5 indicates a low correlation of that explanatory variable with other variables (in practice no-multicollinearity).

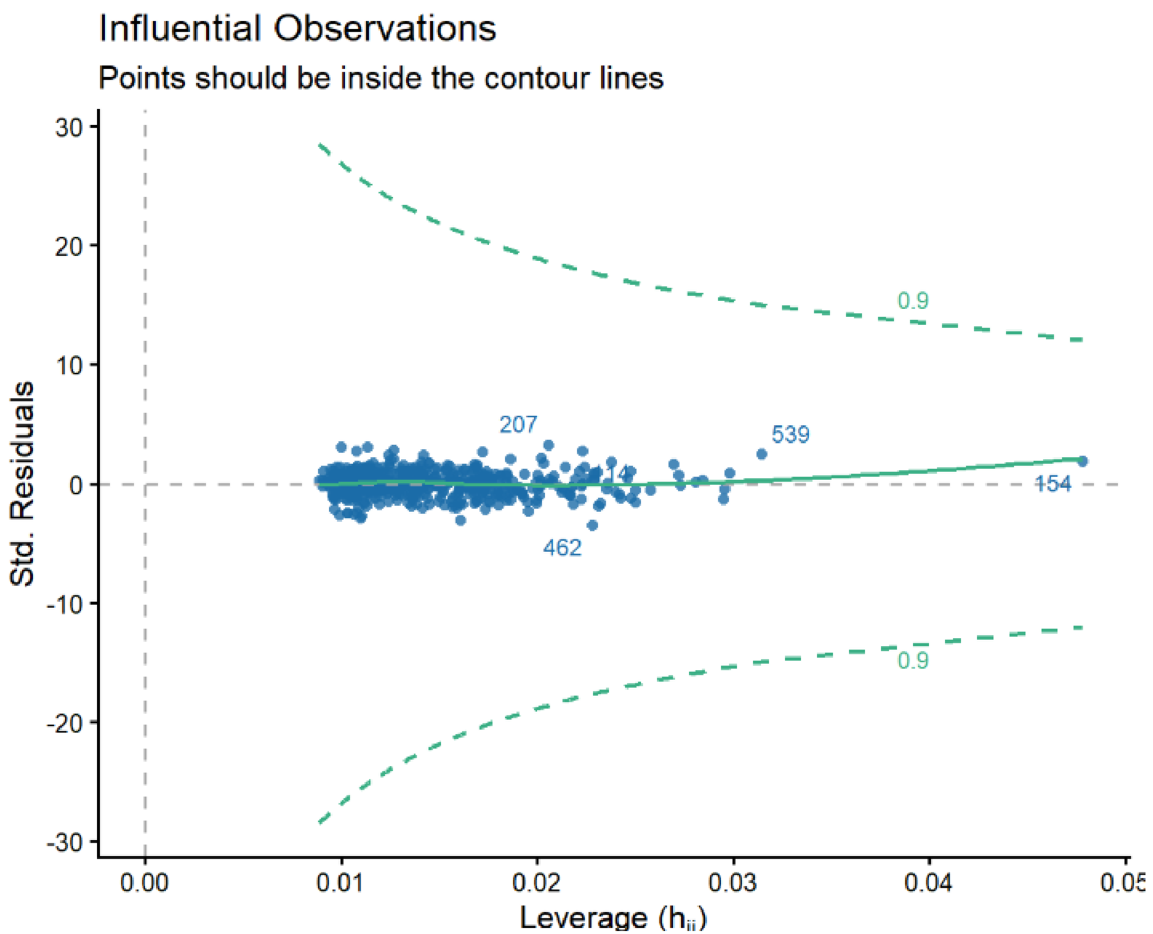
No influential points - Cook's distance

```r
# Calculate Cook's distance
cooks_vals <- cooks.distance(model_full)

# Get Min, Median, Mean, and Max
summary(cooks_vals)
```

```
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
1.000e-08 1.286e-04 7.075e-04 1.932e-03 2.264e-03 3.474e-02
```

Cook's distance is a measure used to identify influential data points in regression analysis. In general, Cook's distances greater than 1 indicate an outlier that may influence the model.



Results table

```
summary(model_full)
```

```
Coefficients:
                          Estimate Std. Error t value Pr(>|t|)    
(Intercept)              -7082.073    500.418 -14.152  < 2e-16 ***
height                     130.789      8.616  15.180  < 2e-16 ***
headc                      109.327     15.604   7.006 7.28e-12 ***
genderMale                 196.142     34.955   5.611 3.21e-08 ***
educationyear12            -34.987     48.421  -0.723   0.4703    
educationtertiary          -36.293     37.446  -0.969   0.3329    
parityOne sibling           75.628     40.630   1.861   0.0632 .  
parity2 or more siblings    95.662     42.042   2.275   0.0233 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 384.6 on 542 degrees of freedom
Multiple R-squared:  0.5967,	Adjusted R-squared:  0.5915 
F-statistic: 114.6 on 7 and 542 DF,  p-value: < 2.2e-16
```

Assessing a model and compare models

Coefficient of determination:

$$R^2 = \frac{explained\ variation}{total\ variation}$$

(R^2 : 0 to 1)

a measure of '**goodness of fit**' of the regression line to the data

Close to 1 ⇨ a large proportion of the variability in the response has been explained by the regression.

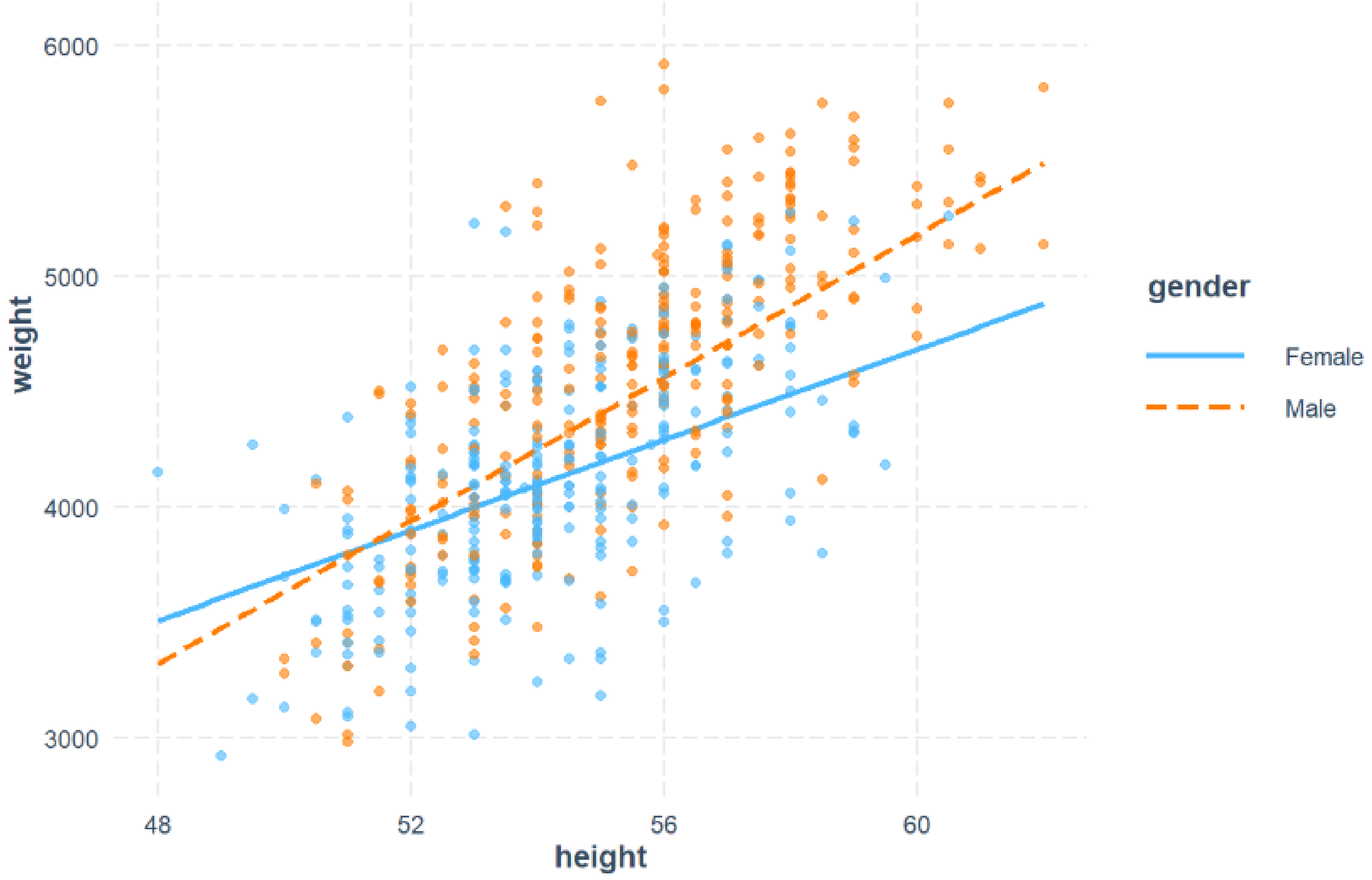
The **adjusted R** square is the R square value adjusted for the number of explanatory variables included in the model. In our example, adjusted $R^2 = 0.59$:
59% of the variation in infant's weight can be explained by the variables in the model.

AIC: compare different models

the smaller value of AIC the better the model

Interaction between height and gender

An interaction term allows the effect of one explanatory variable (e.g., height) on the response (weight) to vary depending on the value of another variable (e.g., gender).



The slopes of the regression lines are not parallel which indicates that there is interaction between height and gender. Here, gender is a moderator.